

Ziad Elhussein

AI / Machine Learning Engineer · Data Science

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SUMMARY

First Class Honours Computer Science graduate (AI and Machine Learning) and Associate Software Consultant at Qestit, testing core-banking systems for Arab African International Bank. Builds ML systems end to end, from data to web app.

EXPERIENCE

Associate Software Consultant · Qestit, Cairo, Egypt Dec 2025 – Present

Client: Arab African International Bank (AAIB) · Temenos T24 core-banking upgrade and cash-management platform

Performance testing · Apache JMeter

- Built and maintained **48 JMeter scripts** for T24 browser, API, instant-payment and ATM channels and three web portals.
- Designed workload models and ran load tests with **1,000+ concurrent users**, including during T24's end-of-day batch.
- Analysed every run (P90/P95 response times, error rates, throughput) and traced each failure to its root cause.
- Engineered a reusable, data-driven test framework that generates test files and times asynchronous file processing.
- Escalated defects to vendors and wrote the formal closure reports for AAIB, with go-live recommendations.

Test automation · Python, Playwright

- Built a **Python Playwright** suite of **1,000+ end-to-end tests** for T24 Accounts, Customers and Funds Transfers.
- Covered positive and negative scenarios and full maker-checker flows (each create test passes its record ID to the authorisation test), running unattended in the **CI pipeline**.

PROJECTS

Pawlytics · Multimodal AI pet-health assistant · graduation project 🔗 github.com/ziadelh/pawlytics

PyTorch · TensorFlow · Hugging Face · Flask · Node.js · MongoDB

- Trained three deep-learning models and combined them in one app: text (Bio_ClinicalBERT, 20 conditions, **63.8%** accuracy), audio (YAMNet, 8 classes, **71.9%**) and image (EfficientNet-B0, 9 conditions, **73.2%**), all on held-out test data.
- Built all three datasets and published them on Kaggle: ~24.8k images from 20+ online sources, 953 audio clips cut from YouTube videos, 51,841 health records split by dog (no leakage).
- Delivered as a web app (Node.js, Express, MongoDB) with a Flask AI service that combines the results into one report.

Evolutionary robotics · Virtual creatures that learn to climb 🔗 github.com/ziadelh/...mountain-climbing-creature

Python · PyBullet · Genetic algorithms

- Genetic algorithm evolving creature bodies and movement in PyBullet: tree-shaped genome, tournament selection.
- Over a 50-generation run, the number of creatures able to climb rose from 2 to 26 per generation.

Traffic car tracking · Detect, track and count cars in traffic video 🔗 github.com/ziadelh/traffic-car-tracking

Python · OpenCV · Kalman filters

- Classical computer vision, no pre-trained detector: frame differencing, background subtraction, a Kalman filter per car.
- Lost cars are re-identified with their old ID, and the car counts match a manual count on both test videos.

More on GitHub: fake-news detection (97%), loan approval, house prices, speech recognition. 🔗 github.com/ziadelh

TECHNICAL SKILLS

AI & ML	PyTorch, TensorFlow/Keras, scikit-learn, Hugging Face Transformers, OpenCV, NLP, computer vision, audio ML
Data	Python, SQL, pandas, NumPy, Matplotlib, Jupyter, MongoDB, MySQL, SQLite
Software	Node.js, Express, Flask, REST APIs, JavaScript, C++, C#, Git, Docker (basics)
Testing	Apache JMeter, Playwright (Python), CI pipelines, performance testing, UI test automation, Temenos T24

EDUCATION

BSc Computer Science · First Class Honours Graduated Sep 2025

University of London · academic direction: Goldsmiths · **Specialisation:** Artificial Intelligence and Machine Learning

Key modules: Machine Learning, Neural Networks, NLP, Data Science, Intelligent Signal Processing, Databases, Algorithms

LANGUAGES

Arabic (native) · **English** (fluent) · **German** (B1)